

Stanford University — Symbolic Systems B.S. (Human-Centered AI concentration)

Program Dossier · Prepared for Eidee Tan & Family · Fall 2027 Entry · Research date: August 15, 2026

Stanford's Symbolic Systems B.S. is the single cleanest degree in the country for Eidee's plan to understand minds and intelligent machines before learning to govern them: one major that *requires* computation, mathematics, probability, philosophy, empirical cognitive science, and linguistics, then focuses through a five-course concentration — and the **Human-Centered Artificial Intelligence (HAI) concentration** is advised by a faculty panel that includes three Stanford Law School professors (Daniel Ho, Mark Lemley, Julian Nyarko). The most important thing to know: **the major itself has no competitive gate** — any admitted Stanford student who takes the gateway course can declare — so the real gate is Stanford admission itself (**about 3.8% admitted, Class of 2029**), and Stanford **requires the SAT or ACT** again. There is no law/policy concentration inside Symbolic Systems; HAI is the closest fit, and this dossier is honest about that gap.

1. The university at a glance

Fact	Detail	Source
Location / type	Stanford, California (San Francisco Bay Area); private research university on the quarter system; 3 of 7 schools award undergraduate degrees	Stanford Facts — Academics
Identity line	"The wind of freedom blows" (Die Luft der Freiheit weht); ~4:1 student-faculty ratio, 65+ majors	Stanford Facts — Academics
Undergraduate enrollment	7,289 (Autumn Quarter 2025)	Stanford Facts
Overall admit rate	~ 3.8% — 60,646 applied, 2,302 admitted (Class of 2029, Fall 2025 entry; per Stanford's Common Data Set 2025-26)	Stanford IRDS Common Data Set; figures via labeled secondary summary, Cosmic College Consulting
Test policy	SAT or ACT required for first-year applicants (self-reported at application; official scores only if enrolling); superscoring accepted	Standardized Testing
SAT middle 50%	1520–1570 (enrolled students, Class of 2029); ACT 34–36	Standardized Testing
Application platform + fee	Common Application; \$100 nonrefundable fee or fee waiver	First-Year Applicants
Early deadline	Restrictive Early Action: November 1 (Oct 15 if submitting an arts portfolio; portfolio materials Oct 20); decisions by mid-December	First-Year Applicants

Fact	Detail	Source
Regular deadline	Regular Decision: January 5 (Dec 5 if submitting an arts portfolio; materials Dec 10); decisions by early April	First-Year Applicants
Financial aid headline	Need-blind for U.S. citizens, permanent residents, undocumented students, and eligible noncitizens (need-aware for other international applicants); full need met; no tuition under \$150K income; no tuition, room, or board under \$100K (typical assets); no loans expected	International Applicants · Affording Stanford

2. The big picture — what kind of place this is

Stanford is a large private research university that behaves, for undergraduates, like a much smaller school: roughly 7,300 undergraduates share a campus with about 10,000 graduate students and a faculty large enough to produce an approximately **4-to-1 student-faculty ratio**. Its intellectual personality is unmistakably interdisciplinary and builder-oriented — the university that sits next to Silicon Valley also houses one of the world’s most influential law schools, and the traffic between the two is constant. Undergraduate degrees come from three schools (Humanities and Sciences; Engineering; Earth, Energy & Environmental Sciences); students apply to the university as a whole, not to a school or major.

Undergraduates study on the quarter system — three ten-week terms a year plus summer — which makes it normal to take many distinct courses and to change direction quickly. General education is real but light-touch: a first-year **COLLEGE (Civic, Liberal, and Global Education) requirement** (two courses now, **expanding to three quarters starting in 2027-28 — Eidee’s entering year**), eleven “Ways of Thinking / Ways of Doing” breadth courses spread across eight skill areas, a two-level writing requirement, and one year of a foreign language. None of it is a prescribed canon; the major provides the intellectual spine.

For Eidee’s cognition→AI→law direction, Stanford is arguably the archetype school. Symbolic Systems — the interdisciplinary major that studies minds and machines together — is a Stanford invention whose alumni populate both AI labs and technology policy. The university layered onto it the **Stanford Institute for Human-Centered Artificial Intelligence (HAI)**, founded in 2019 under Fei-Fei Li and James Landay, and the law school runs two research centers squarely on Eidee’s target: **CodeX**, the **Stanford Center for Legal Informatics** (computational law) and the **RegLab** (machine learning for government and regulation), led by faculty who — unusually — also advise undergraduates in the Symbolic Systems HAI concentration.

Music sits comfortably inside campus life rather than in a conservatory silo. The Department of Music states plainly that “**participation in all instrumental ensembles, lessons, and activities is available by audition to students in any major**”, and it hosts over 150 concerts a year. A serious violinist majoring in Symbolic Systems is a normal Stanford profile, not an exception.

3. The program — Symbolic Systems B.S., Human-Centered AI concentration

Exact name and home: Bachelor of Science in Symbolic Systems, an interdisciplinary program housed in the **School of Humanities and Sciences** (**major policies and requirements**). The major is built for students who want to study intelligence — human and artificial — with real technical tools, and it is the one degree here where Eidee’s breadth (math, computing, psychology, philosophy, language) is the requirement, not a side interest.

The core (G4 requirements, in effect since Fall 2020)

Per the program's [core requirements page](#), every Symbolic Systems major completes:

Component	What it is	Rough count
Gateway	SYMSYS 1 "Minds and Machines" — required before declaring	1 course
Mathematics	Single-variable calculus (MATH 19–21) + multivariable/linear algebra (e.g., MATH 51)	≥14 units
Philosophical Analysis	Intro philosophy; PHIL 80 (writing in the major); one advanced philosophy course (e.g., PHIL 181 Philosophy of Language, PHIL 186 Philosophy of Mind—area options)	3 courses
Formal Methods	Formal logic (PHIL 150 / CS 157); theory of computation (CS 103 / CS 154); probability & statistics (e.g., CS 109, STATS 116)	3 courses
Computational Methods	Programming (CS 106A → CS 106B) plus one post-106B course (options include CS 221 AI, CS 229 Machine Learning, CS 147 HCI)	3 courses
Empirical Cognitive Science	PSYCH 1 ; an area course in cognition/language/neuroscience (e.g., PSYCH 45 Learning & Memory, LINGUIST 130A); a linguistic theory course	3 courses
Cross-Area	1 course combining methods of two or more areas (e.g., PSYCH 204, PHIL 152)	1 course
Advanced Small Seminar	Discussion seminar of ≤20 students building on the core	1 course
Capstone	A practicum (project/research course such as SYMSYS 190 or CS 194-style options) + an integrative course, begun junior year or later	2 courses

Core courses must be taken for a letter grade with **C- or better**, and every course counted toward the major must be **3+ units** ([major policies](#)).

The concentration layer — and HAI specifically

On top of the core, every major completes a **five-course concentration**. There are currently **16 named concentrations** — Artificial Intelligence, Cognitive Science, Human-Computer Interaction, Natural Language, Neurosciences, Decision Making & Rationality, Computer Music, and others — plus an Individually Designed Concentration option (Section 7 covers its limits).

The **Human-Centered Artificial Intelligence (HAI) concentration** is the one built for Eidee's thesis. Its five courses come one each from five areas (per the program's official requirements document linked from that page):

- Digital Technology Ethics and Policy** — e.g., **CS 182 Ethics, Public Policy, and Technological Change**, CS 181 Computers, Ethics, and Public Policy, CS 120 Introduction to AI Safety, CS 281 Ethics of Artificial Intelligence.
- Human Impact** — e.g., **CS 209 Law, Order & Algorithms, LAW 4039 Regulating Artificial Intelligence**, COMM 154 The Politics of Algorithms, SYMSYS 201 Digital Technology, Society, and Democracy. Actual law-school and law-adjacent courses live here.
- Augmenting Human Capabilities** — e.g., CS 147 Introduction to Human-Computer Interaction Design, CS 247A Design for Artificial Intelligence, CS 278 Social Computing.
- Intelligence** — the technical AI pillar: CS 221 Artificial Intelligence, CS 229 Machine Learning, CS 224N Natural Language Processing with Deep Learning, CS 231N Deep Learning for Computer Vision.

5. **Integrative requirement** — a synthesizing course taken junior year or later (options include CS 281, PSYCH 202, SYMSYS 202).

The advisor lineup is the quiet headline. The concentration's recommended advisors, listed on the [official concentration page](#), include **three Stanford Law School professors — Daniel Ho (RegLab founder), Mark Lemley (IP law), and Julian Nyarko (computational law)** — alongside computer scientists (Michael Bernstein, Emma Brunskill, Chelsea Finn, James Landay), and faculty from education, the business school, linguistics, history, and music. An undergraduate concentrating in HAI chooses a concentration advisor from this list: a structural, named channel to exactly the AI-law faculty Eidee wants to learn from. (Advising is real; it is not a promise of research placement.)

Signature course

CS 182 / ETHICSOC 182 / COMM 180 / PHIL 82 / POLISCI 182: Ethics, Public Policy, and Technological Change is Stanford's flagship technology-ethics course, cross-listed across five departments and [taught through units on algorithmic bias, data privacy, AI, and platform power](#) with philosophy, policy, social science, and engineering lenses (prerequisite: CS 106A). It satisfies the HAI concentration's ethics-and-policy pillar and is the single course that most compactly embodies Eidee's whole direction.

Capstone, honors, and coherence

The capstone's two-course arc (practicum + integrative) forces synthesis rather than course-collecting. Ambitious students add the **Symbolic Systems Honors Program**: open to declared majors with junior/senior standing, a **3.0+ GPA, and an accepted proposal** (due by Week 2 of Autumn Quarter for same-year completion), it runs through SYMSYS 190/191, requires a faculty advisor plus second reader, spans multiple quarters, and ends in a public presentation at the SymSys Forum. An honors thesis on, say, cognitive foundations of AI regulation — advised by an HAI-list law professor — is the natural summit of this pathway.

What makes the structure coherent rather than a pile of electives: the core guarantees that Eidee can *do the math, write the code, run the experiment, and make the argument*; the HAI concentration then points all of that at one question — how intelligent systems should serve human beings — with the ethics/policy pillar built in, not bolted on.

4. How the pieces fit together

Entry unit. You apply to Stanford University, not to a major or school. Nothing about intended major is binding.

Declaration. Symbolic Systems is **not an impacted or capped major**. Per the program's [declaration page](#), a student needs one quarter of completed coursework and SYMSYS 1 (taken or in progress past the drop deadline), then declares in Axess, files a course plan, and holds a short declaration interview with a peer Advising Fellow — no application, no GPA screen, no cap. Most students on the honors track declare in the first or second year (the program publishes [sample schedules](#) for early and late declarers).

Double-counting. Concentration courses may double-count with core requirements under specified conditions, with substitute electives sometimes required; the final course plan must be approved **at least two full quarters before graduation** ([major policies](#)). The advanced small seminar and capstone can sometimes be satisfied through concentration coursework.

Scheduling reality. The quarter system makes the ~19-course major (core + concentration) fit comfortably in four years alongside GE, orchestra, and lessons — the published sample schedules show honors students carrying philosophy, CS, and

psychology simultaneously from year one. The practical constraint is not unit count but sequencing: CS 106A → 106B → CS 109/103 and MATH 19–51 should start early because upper-level AI courses (CS 221/229/224N) sit behind them.

Gates beyond university admission - Stanford admission — competitive (~3.8%, Class of 2029). The only truly selective gate on this pathway. - **SymSys declaration** — requirements-based (one quarter + SYMSYS 1 + advising interview); uncapped. - **HAI concentration** — requirements-based (course plan approval); no application. - **Individually Designed Concentration** (if ever used) — requirements-based but needs three approvals (advisor, Associate Director, Director) ≥2 quarters before graduation. - **Honors program** — requirements-based (3.0 GPA + accepted proposal + faculty readers). - **Private violin lessons & ensembles** — **audition required** (open to all majors). - **Research positions (HAI institute, CodeX, RegLab, CS/Psych labs)** — competitive/selective; student-initiated; never guaranteed.

5. An illustrative four-year path

*Illustrative only — not official advising. Modeled on the program’s published **sample schedule** for an early declarer with honors.*

Year	Autumn / Winter / Spring (quarter system, ~4 courses each)
Year 1	SYMSYS 1 (autumn); CS 106A→106B; MATH 19/20/21→51; PSYCH 1; intro philosophy; COLLEGE requirement (three quarters for the class entering 2027); PWR 1; Stanford Symphony Orchestra audition in week 1; lesson-studio audition. Declare SymSys as early as winter/spring.
Year 2	CS 103, CS 109; PHIL 80; PHIL 150 or CS 157; cognition area course (e.g., PSYCH 45); linguistics theory course; CS 182 (Ethics, Public Policy & Technological Change) as the HAI ethics/policy pillar; continue orchestra + lessons.
Year 3	CS 221 or CS 229 (Intelligence pillar, also post-106B core course); CS 147 (Augmenting Human Capabilities); CS 209 Law, Order & Algorithms or SYMSYS 201 (Human Impact); cross-area course; advanced small seminar; choose HAI concentration advisor (e.g., a RegLab/CodeX-affiliated law professor); begin research outreach; capstone practicum can start spring.
Year 4	Integrative requirement; honors thesis via SYMSYS 190/191 (proposal by Week 2 of autumn) on an AI-governance or human-AI cognition question; SymSys Forum presentation in spring; finish Ways breadth; continue chamber music (MUSIC 171).

6. Beyond the classroom

Stanford HAI (the institute). Founded 2019, **HAI** runs research, policy, and education programs on human-centered AI. For undergraduates the **named channel** is precisely the SymSys HAI concentration, plus student affinity groups and courses taught by HAI faculty; its formal fellowship program is graduate-level. Treat the institute as an ecosystem to plug into (talks, groups, faculty), not a program that admits undergraduates.

CodeX — Stanford Center for Legal Informatics. A **Stanford Law School–Computer Science partnership** on computational law, legal analytics, and augmented legal practice. Its FutureLaw week includes an **LLM x Law Hackathon** and an agentic-AI bootcamp explicitly for “interdisciplinary Stanford student teams.” Honest note: CodeX publishes graduate fellowships and courses; it does **not** advertise a formal undergraduate research program — undergrad involvement happens through events, hackathons, and individually arranged work.

RegLab. Daniel Ho’s **Regulation, Evaluation, and Governance Lab** builds machine-learning tools with government agencies (compliance, adjudication, policy evaluation). Its site says “students, data scientists, engineers... are encouraged to reach out” — an open door, not a program; positions are selective and student-initiated. Because Ho is a listed HAI-concentration advisor, the concentration is the most realistic introduction path.

The broader ethics/policy ecosystem. The McCoy Family Center for Ethics in Society co-runs the Embedded EthiCS program with computer science and HAI; the Public Policy program hosts CS 182; philosophy of mind/language and linguistics are core departments for the major, not outside electives.

Violin and music life. For an advanced violinist who is not a music major: the **Stanford Symphony Orchestra (MUSIC 160)**, **Stanford Philharmonia (160A)**, **Stanford Baroque Soloists (160C)**, and **Chamber Music (MUSIC 171)** are all **open by audition to students in any major**. **Private lessons** (violin faculty include Debra Fong) require a studio audition and enroll as regular Axess courses; for non-majors the published fee is **\$595 per studio per quarter (2025-26)**, and undergraduates on aid may request lesson support **through the Financial Aid Office** — worth confirming amounts directly, as the department does not publish subsidy levels. In admission, an **optional arts portfolio** in music (reviewed by fine-arts faculty at the admission office’s discretion) is available — but it moves the whole application deadline earlier (see Section 8).

7. Access, feasibility, and honest risks

The honest gap: there is no law, policy, or “AI governance” concentration in Symbolic Systems. The **16 concentrations** run from AI to Neurosciences, but none is named for law or policy. HAI is the closest built path, and it genuinely contains the material (CS 182, CS 209, LAW 4039) and the law-faculty advisors — but law remains one pillar of five, not the organizing theme. The **Individually Designed Concentration** could in principle be built around “computational law and cognition,” yet it has real limits: five courses with a stated coherent theme, **at least two addressing SymSys core breadth areas**, one satisfying the integrative requirement, and sign-off from the advisor, Associate Director, and Director **no less than two full quarters before graduation** — miss that window and you complete a standard concentration or delay graduation. Practical read: plan on HAI, and treat an IDC as a possible refinement, not the plan.

Flexibility can become diffusion. SymSys’s breadth is its glory and its trap; without an early through-line (the seed research called it “the spine, not a label”), four years can pass in excellent but disconnected courses. The concentration advisor and the honors thesis are the antidotes — use both.

Access realities. Popular CS courses are large; seats in specific quarters, capstone topics, and research mentorships are never guaranteed. HAI-institute fellowships are for graduate students; CodeX and RegLab involvement is real but student-initiated and selective. Private lessons carry a per-quarter fee (aid help possible but not published as a guarantee). None of this blocks the pathway; all of it requires initiative.

Admission is the gate. With a ~3.8% admit rate (Class of 2029) and an SAT middle-50% of 1520–1570, no applicant should treat this school as likely. If admitted elsewhere but not here, the closest analogues in Eidee’s set carry more structure per dollar of luck.

When a different Stanford pathway would be better. If Eidee decides she wants maximal engineering depth, the CS B.S. (AI track) with SymSys-style electives is the honest spine; if governance overtakes cognition as the primary interest, Public Policy with a computation cluster would fit better. SymSys-HAI is the right choice precisely while “understand intelligence first, govern it second” remains the thesis.

8. Applying — logistics for Fall 2027

Stanford accepts the **Common Application** with a **\$100 fee or fee waiver**; two teacher recommendations plus a counselor letter are required (**first-year applicants**). Deadlines below are the university's current published cycle (page updated July 2026); confirm in August 2026 when the 2026-27 application opens.

Item	Date	Notes
Restrictive Early Action	November 1, 2026	Non-binding but restrictive: no other private-college EA/REA/ED/EN and no binding public ED (REA rules); decision by mid-December
REA with arts portfolio	Application October 15; portfolio materials October 20, 2026	Portfolio pulls the entire application 2+ weeks earlier (arts portfolio)
Regular Decision	January 5, 2027	Decision by early April
RD with arts portfolio	Application December 5; materials December 10, 2026	One month before the standard RD deadline
Testing (required)	REA: ACT by end of September, SAT by end of October 2026; RD: either by end of December 2026	Testing policy ; superscores accepted — Eidee's 1510 sits just below the enrolled 25th percentile (1520, Class of 2029), so a retake is worth considering
Interview	Optional alumni conversation (~40 min, in person or video); "complete with or without an interview"	Interview policy
Financial aid	REA: CSS Profile + IDOC by November 15, 2026 (aid estimate with mid-December decision); RD: by February 15, 2027 ; FAFSA per checklist for U.S. applicants	Aid requirements and dates

Decision point for the family: the **arts portfolio is optional** and reviewed at the admission office's discretion. If Eidee's violin recording is truly distinguished, it can be submitted — but it costs 2–4 weeks of application time at each round. Since violin is her super-curricular rather than her intended field, submitting at RD (Dec 5) or skipping the portfolio in favor of strong essays about violin-as-thinking are both legitimate strategies.

9. Sources

Admissions: First-year applicants (deadlines, fee, platform) · Standardized testing · REA/RD decision process · Interview · International applicants / need-blind statement · Stanford IRDS Common Data Set · CDS 2025-26 summary (secondary, labeled) · Stanford Facts · Stanford Facts — Academics

Program: SymSys major policies & requirements · Core requirements · Concentrations list · HAI concentration (incl. advisor list & requirements doc) · Individually Designed Concentrations · How to declare · Honors program · Sample program schedule · CS 182 course page (Public Policy program) · Stanford HAI — about · HAI for Stanford students · CodeX · RegLab · GE requirements (bulletin)

Music: Instrumental performance & ensembles · Lessons · Lesson funding & fees · Arts portfolio

Aid: Affording Stanford · Parent contribution / income thresholds · Aid application requirements & dates